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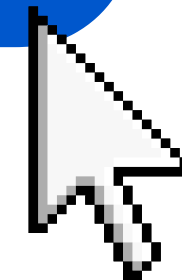
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Group 24

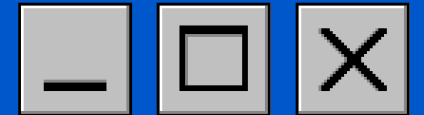
FUELGUARD

FUEL DISTRIBUTION AND
ALLOCATION



Start





PROJECT OVERVIEW



What will the project do?

FuelGuard records and verifies fuel shipments (petrol/diesel) from import terminals to retail stations, ensuring fair allocation during shortages.

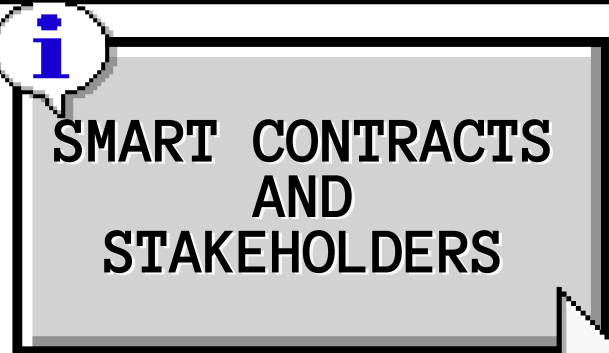
How will you do it?

- Store fuel batch data (importer, fuel type, volume, date, location) as on-chain records via smart contracts.
- Track ownership transfers: Importer → Wholesaler → Retailer.
- Provide read-only verification for regulators and the public.

Why is it important?

- Exposes hoarding and unfair prioritisation.
- Builds trust in rural areas and reduces panic buying.
- Gives regulators (ACCC) auditable, immutable evidence.

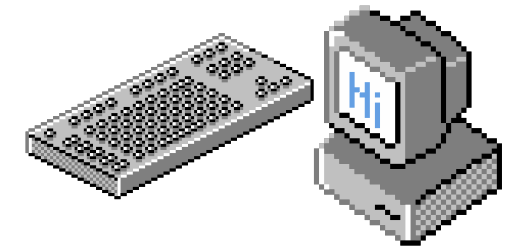
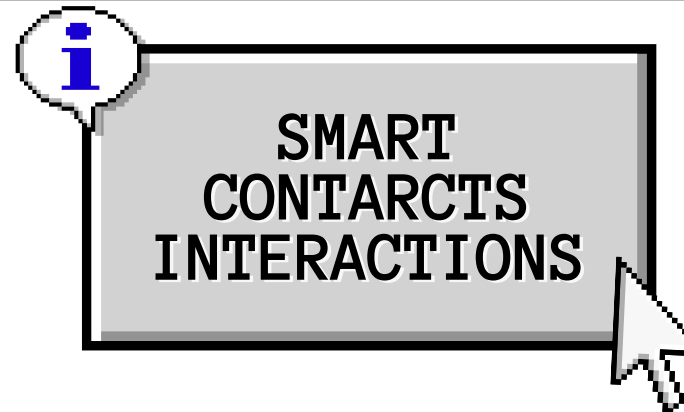




Record: `recordFuelBatch()`-(creates a new fuel batch record with unique ID)
→ Importers: `write(record arrivals)`

Allocation: `allocateToWholesaler()`, `distributeToRetailer()`, `confirmDelivery()`
→ Wholesalers: `write (transfer to retailer)`, Retailers: `write (confirm delivery)`

Verification: `getBatchHistory()`, `getCurrentHoldings()`
→ Regulators (ACCC): `read all`, Public/Consumers: `read simplified data`

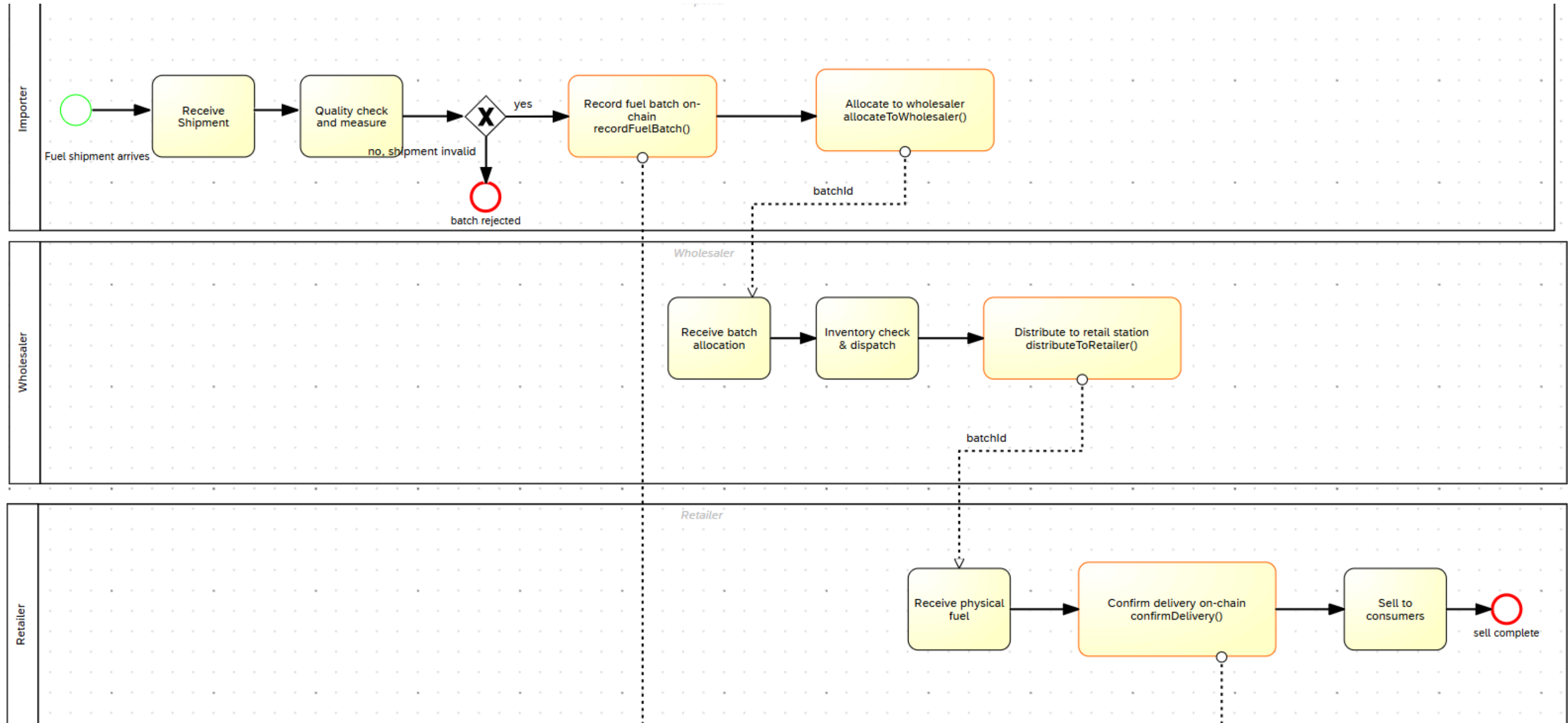


Interaction flow:

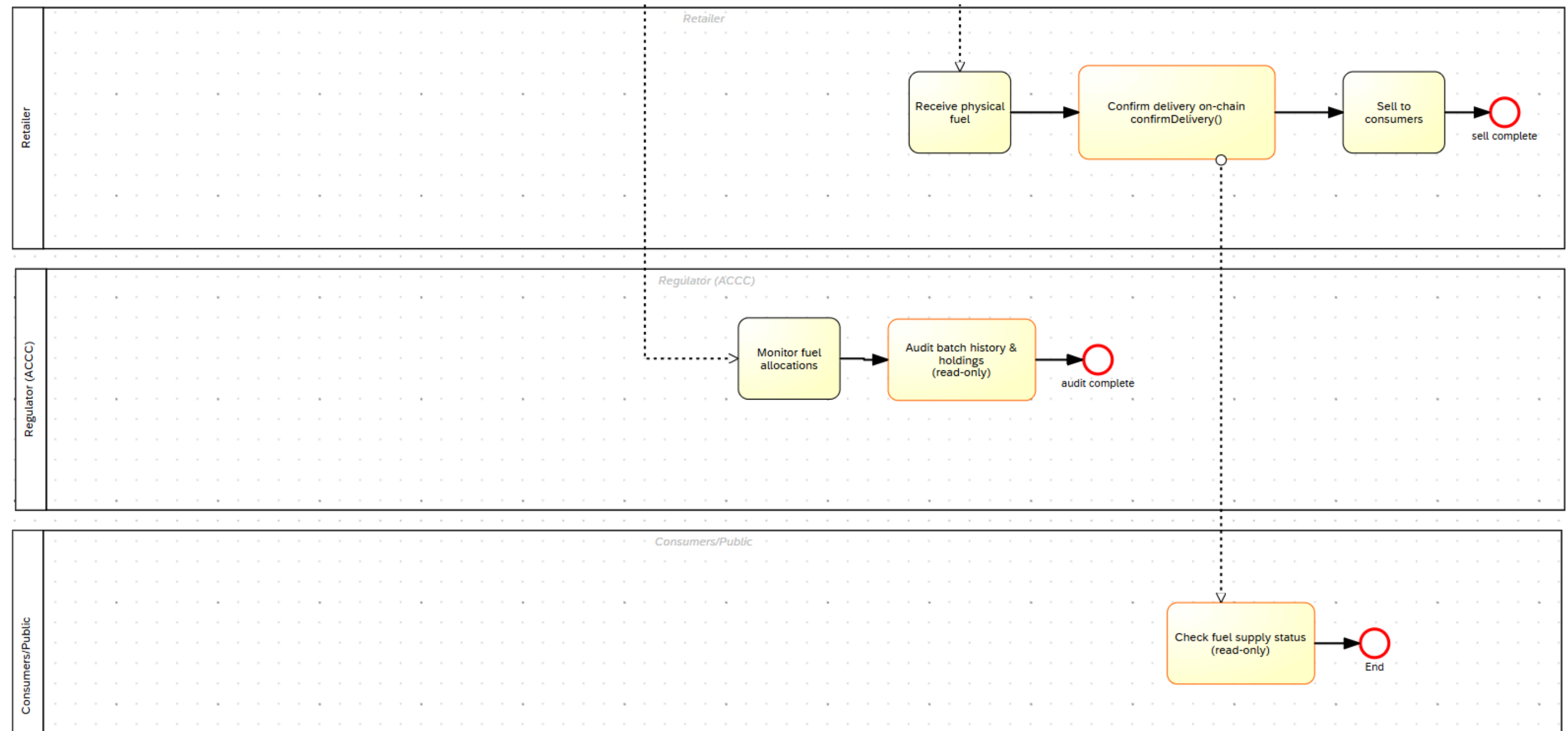
- Importer calls `recordFuelBatch()` on Record contract → new batch record created.
- Importer calls `allocateToWholesaler(batchId, wholesaler)` on Allocation contract → custody is transferred.
- Wholesaler calls `distributeToRetailer(batchId, retailer)` → batch moves to retailer.
- Retailer calls `confirmDelivery(batchId)` → final confirmation recorded.
- Regulator/Public via the Verification contract reads events and current state via getter functions `getBatchHistory()`, `getCurrentHoldings()`- no modifications.

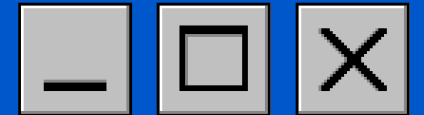


BUSINESS PROCESS MODEL SPECIFICATION



BUSINESS PROCESS MODEL SPECIFICATION





KEY CHALLENGES AND HOW WE ADDRESS THEM

Transparency vs Commercial Security:

Challenge: Public must see fairness; importer want to protect commercial inventory data.

Approach: Tiered access in Verification contract - regulators see detailed volumes, public sees regional data only.

Integrated Regulations (ACCC):

Challenge: Design must meet potential future requirements such as mandatory ACCC reporting or emergency quota systems.

Approach: Read-only Verification contract gives ACCC direct audit access; design extensible for future quota enforcement.

Trust at high-reliability write points:

Challenge: Key write points need trust: import verification, wholesale allocation, retail delivery confirmation.

Approach: Multi-signature approval for important transactions, eg: recordFuelBatch requires both importer and port surveyor signatures.

Honest limitations:

Challenge: 3 constraints we acknowledge upfront in our proposal.

- Supply: can not solve geopolitical fuel shortage.
- Adoption: only valuable with enough stakeholder participation.
- Oracle: off-chain data corruption could poison the ledger



